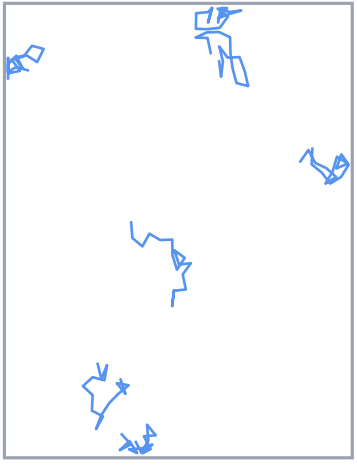
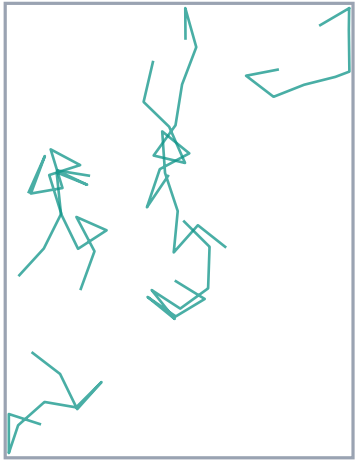
Knudsen number $\mathrm{Kn} = \lambda / L$: how many collisions a molecule has before it crosses the device

continuum
$\mathrm{Kn} < 0.01$



Euler / Navier-Stokes,
no-slip, Fick, Fourier

slip / rarefied
$0.01 - 0.1$

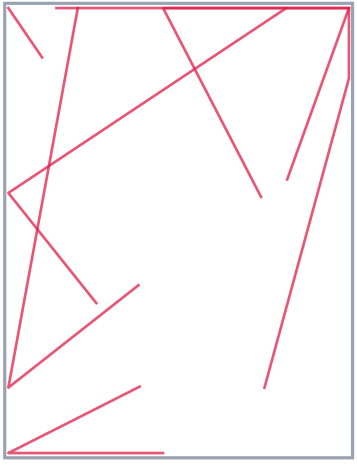


NS + slip & temperature
jump boundary conditions

transition
$0.1 - 10$

no hydrodynamic closure;
DSMC, moment methods

free molecular
$\mathrm{Kn} > 10$



ballistic molecules,
Poisson noise